

Bottleneck-Mamba 3D U-Net Baseline for Lesion Segmentation in ISLES'26 Ischemic Stroke: Limited by Precision in Segmenting Small and Multifocal Lesions

Konark Karia¹ and Megh Patel¹

¹ MTech CSE, SEAS, Ahmedabad University
{konark.k@ahduni.edu.in, megh.p@ahduni.edu.in}

Abstract. Here, we introduce our first baseline method for the ISLES'26 competition, which requires participants to segment ischemic stroke lesions in the native space using T1-weighted MRIs in acute, sub-acute, and chronic phases. Instead of employing the state-of-the-art global context transformer block used by most current segmentation networks, we use a standard encoder-decoder architecture that incorporates a bidirectional state-space (Mamba) layer at the bottleneck in place of the global context block, an approach that scales linearly with sequence length as opposed to the quadratic scaling of the global context block. Trained on a single train/validation split taken from the ISLES'26 training dataset (1235 train/218 validation out of a total of 1453 cases) with the use of a combined Dice and cross-entropy loss, the proposed network achieves a validation Dice score of 0.501. Looking into the performance per case shows a different picture than just the overall score: while the network seldom misses any lesions, it tends to over-segment them. The mean precision in the validation set is at 0.43 vs the mean recall at 0.61. For lesions below the dataset median size, this gap almost triples: 0.21 precision vs 0.48 recall. In close to a third of the lesion-positive cases, the lesion volume is over-estimated at least threefold. We consider the precision-recall gap above to be the key point of our submission and will describe the loss re-weighting and post-processing steps that will be done in the camera-ready version.

Keywords: ischemic stroke, lesion segmentation, state space models, Mamba, 3D U-Net, magnetic resonance imaging, ISLES 2026

1 Introduction

Tracing the infarction region on the brain due to ischemia via MRI continues to be a tedious process, even though it is still a common process in stroke research and treatment, and the manual tracing of these infarctions has proved to be unable to cope with the huge volume of data being collected as part of multi-centre studies. This has been an issue in the different ISLES challenges that have been conducted. The most recent one, ISLES'22-ATLAS, has brought into light the generalization gap which has been experienced. The models trained on only one acquisition protocol failed when it came to inferring from the unseen centres' data. ISLES'26 is trying to address this

problem using almost 2000 annotated T1-weighted images from over 60 centres around the world.

First, convolutional encoder-decoder architectures in the family of nnU-Net have been consistently doing well due to the way in which the framework configures itself automatically based on the given dataset [1]. Transformers such as UNETR and Swin UNETR partially substitute the convolutional backbone with self-attention which increases the receptive field significantly, much beyond the capabilities of a stacked set of convolutions, although the downside is the increased computational costs of quadratic order with respect to the length of the sequence [2]. Finally, recent innovations in the area include state space sequence models. In this architecture family, the S4 state space models are reengineered in the input-dependent or selective recurrence combined with a hardware-aware parallel scan; these models perform comparable to or even better than attention-based models with the advantage of scaling linearly [3]. Vision Mamba extended the idea to the case of image data by adding a bidirectional scan, which was needed since visual information lacks an intrinsic orientation in contrast to text [4]. The first application to biomedical segmentation involved a redesign of Mamba blocks inside an nnU-Net encoder-decoder with two proposed architectures – Enc, where each encoding layer is followed by Mamba blocks and Bot, where there are Mamba blocks at the bottleneck only [5]. SegMamba brought the innovation further by introducing a tri-directional scan for volumetric data [6].

With respect to the computational budget allocated for this assignment, which consists of one GPU of 8GB capacity, it makes sense to try out the lightweight “U-Mamba Bot” variant for the start and save the more complex designs for further use. The task for us at this stage is only to obtain a working prototype, figure out exactly how it works badly on this particular dataset, and use this failure analysis to determine our next move.

2 Methods

2.1 Data

The training dataset released to the challenge participants was comprised of 1453 T1-weighted native-space images each accompanied by a binary lesion mask from the combined multi-center pool described in the challenge introduction (mostly based on the ATLAS lesion-tracing database [7] but augmented with other contributing datasets). Images are pre-segmented (skull-stripped) and isotropic (approaching 1mm) in space, though individual dimensions may vary slightly within 150-190 voxels range per axis. Sixty-six percent of images contain more than one lesion component and lesion volume distribution is strongly right-skewed, with a median volume of about 5,400mm³ due to several cases with significantly larger lesions.

A set of 218 (15%) cases was reserved for validation and kept fixed throughout the challenge (seeded only once before training), leaving 1235 cases for training. This is a single split rather than k-fold cross-validation, which is considered an acceptable strategy for internal validation in this challenge; k-fold cross-validation is one of the following steps described in Section 4.

2.2 Preprocessing and sampling

Each volume was normalized using z-score intensity normalization on brain voxels only. No resampling in space was done as the sampling was nearly isotropic in all of our subjects. Patches of $96 \times 96 \times 96$ voxels were sampled directly from native resolution images. Because of the small size of lesions compared to a $96 \times 96 \times 96$ patch and multifocality of the data set, we used foreground-weighted patch sampling, where 66% of the time, a patch was centered on a randomly chosen lesion voxel, while the rest of the time a patch was sampled from any random location in the brain.

2.3 Architecture

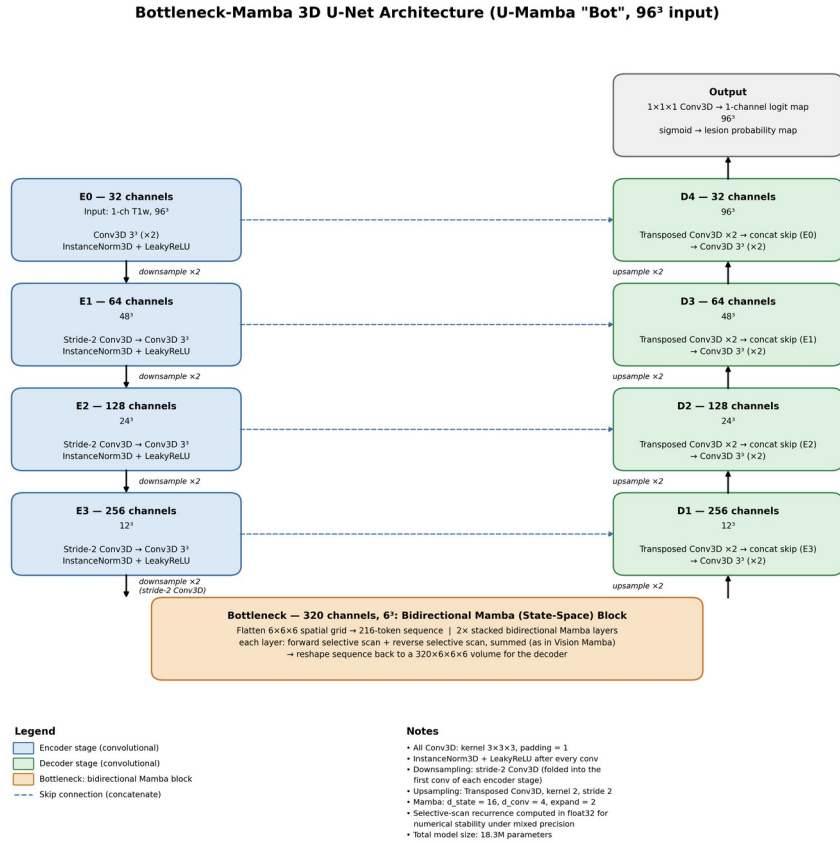


Fig. 1. Architecture of the proposed bottleneck-Mamba 3D U-Net (the U-Mamba “Bot” variant). The encoder (left) and decoder (right) are ordinary 3D convolutional stages joined by skip connections at matching resolutions; the state-space mechanism is confined to the bottleneck, where the flattened $6 \times 6 \times 6$ feature map is processed by two stacked bidirectional Mamba layers before being reshaped back into a volume for the decoder.

The network architecture is a 5-layer convolutional encoder-decoder with channels from 32 to 320 (32-64-128-256-320), instance normalization, LeakyReLU activation functions, and U-Net-like skip connections, which looks somewhat similar to architectures of both nnU-Net and U-Mamba [1,5]. In the bottleneck layer, 6x6x6 spatial feature map is reshaped to 216-token sequence and processed by two Mamba layers in bidirectional manner, where only a single selective scan recurrence is run in each direction and summed as done in Vision Mamba [4]. Finally, the output vector is reshaped back to volume representation for subsequent decoding process. Our whole model contains 18.3M parameters.

Unfortunately, because of the absence of the Mamba selective scan CUDA kernel in our training setup, we had to implement the selective scan recurrence manually in PyTorch, running it in float32 precision to maintain numerical stability of the mixed-precision training regime. Our implementation is equivalent to the fused one, however, it works much slower per epoch, and performance overhead limits the number of epochs which could be trained within the timeline of this submission.

Figure 1 provides the complete stage-wise architecture of the network. In the encoder, each layer runs two 3x3x3 convolutions, the first one being responsible for applying the stride when performing the downsampling, followed by instance normalization and a LeakyReLU activation; after four such downsampling steps, the 96³ initial volume becomes the 6³ bottleneck volume with the channel size being doubled at each step from 32 up to 320. On the other hand, the decoder performs the opposite procedure: stride-2 transposed convolution upsample the feature map, concatenates the corresponding encoder stage output and applies the same two-convolution block. Two bidirectional Mamba layers at the bottleneck use the following parameters: state dimension of 16, convolution width of 4, and expansion factor of 2. At the end of the decoder part, the 1x1x1 convolution reduces the feature map with 32 channels to the logit, which is converted to lesion probability map by applying sigmoid activation.

2.4 Training

Training used a combined Dice and binary cross-entropy loss with AdamW, a peak learning rate of 2e-4 under cosine decay with linear warmup, automatic mixed precision, and gradient clipping. To fit within 8GB of GPU memory, batch size was kept to 2, with gradient accumulation over 5 steps producing an effective batch of 10. Validation ran every 5 epochs via Gaussian-weighted sliding-window inference over full volumes at 50% overlap. We report results from two training stages: an initial 300-epoch run, followed by a second run that warm-started from those weights, applied the augmentation described above, and trained for a further 100 epochs under a fresh learning-rate schedule.

3 Results

The initial run reached a best validation Dice of 0.4585. Warm-starting from those weights, adding augmentation, and running a fresh 100-epoch schedule pushed this to

0.5010. Both represent an early checkpoint in what is still an ongoing optimization process, and training loss was still declining slowly, if slightly, by the end of the second run.

To understand what stood between these numbers and a genuinely strong result, we computed per-case precision and recall alongside Dice across the 218-case validation split (Table 1). Mean recall (0.613) comfortably exceeds mean precision (0.433), which tells us the model is not primarily struggling to locate lesions. It is predicting more lesion tissue than actually exists. That gap widens considerably once lesion size enters the picture.

Table 1. Validation-set precision/recall by lesion size (n=218; two cases have zero ground-truth lesion volume and are excluded from the size split).

Group	n	Mean Dice	Mean Precision	Mean Recall
All cases	218	0.459	0.433	0.613
Below-median lesion size	108	0.258	0.214	0.478
Above-median lesion size	108	0.668	0.658	0.748

In total 216 of 218 validation cases had a nonzero lesion in the ground-truth image. For them, the median ratio of the predicted and true lesion volume is relatively modest with the value of 1.53. However, the mean is extremely high – 17.4 due to a set of severely over-predicted cases. In particular, 31% of lesion-positive cases are over-segmented more than three times while 8.8% are over-segmented more than ten times. At the same time, under-prediction (when predicted volume is less than half of the true volume) occurred in 5.6% of cases. Remarkably, in two of the validation cases without lesions the network predicted a nonzero lesion volume.

There were two validation cases with the lesion volumes several orders of magnitude greater than the rest of the dataset ($\sim 10^6$ voxels vs median of several thousands). We have not decided yet whether it is a label noise associated with a particular participating center or a special spacing convention that we are not currently aware of.

4 Discussion and Conclusion

Combined Dice and cross-entropy loss does not really penalize a false positive blob in close proximity to a small lesion very strongly, as the volume of (the small) true lesion in its denominator is already very high relative to even a false positive blob added, in contrast to the much smaller false positive blob relative to a much higher denominator representing a large lesion. With 66% of ISLES’26 cases featuring multifocal disease and with the distribution of lesion sizes biased heavily towards small satellite lesions, this would constitute a pretty much worst case scenario for this particular loss formulation. The next step would seem natural which was switching to Tversky loss [9] to favor stronger penalty for false positives (with high alpha and low beta) and post-processing for connected component analysis yet by the time we made our submission, we have not tested it yet. Given that the median over-segmentation is moderate 1.5x

while mean is 17.4x, the conclusion is clear, a few false predicted components do it rather than general presence of noise in all predictions.

There are a few limitations that affect the generality of the presented results. First, we employed only one randomly selected train/validation split without cross-validation as required by the workshop rules for internal validation of our model. While this approach is acceptable under the workshop rules, it provides less accurate estimates compared to aggregating them between splits. Additionally, we employed only the Bottleneck variant (“Bot”) of Mamba, but there was also the “Encoder” variant that utilized encoder’s state space, and SegMamba with tri-directional scan that potentially could handle the issue with multifocal lesions better, because they provide the state-space mechanism access to context before any details are lost due to downsampling. Lastly, we should directly evaluate whether there is an issue of different voxel spacing in contributing datasets with regard to anomalous lesion-volume cases presented in Section 3.

This submission is purposely partial, something that the call explicitly allows, due to the fact that the algorithms are not yet fully developed. Between now and the camera ready date, we intend to submit results of our experiments involving the precision weighted loss and post processing changes mentioned above, and, time permitting, also a comparison against nnU-Net and Swin UNETR with identical training splits. This will give an indication of whether the differences that we have seen here come from architecture, loss, or data itself.

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